

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination 2024-25 (NEP)

DEPARTMENT OF COMPUTER SCIENCE**Paper Code: CSCMJ11 / CSCMN11 — Introduction to Programming using C**

Time: 2:30 Hours

Maximum Marks: 70

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Answer any five questions, including Question No. 1, which is compulsory.***Question 1** (Compulsory — Attempt any nine)

[9 × 2 = 18]

- (a) Differentiate between `int` and `float` data types.
- (b) List the rules for naming a variable in C.
- (c) What is the role of `continue` keyword in C programming?
- (d) Is it possible to rename a data type in C? If yes, then how?
- (e) What is the use of `sizeof()` in C programming?
- (f) What is the difference between `scanf()` and `fscanf()` functions?
- (g) Write the syntax and usage of ternary operator in C.
- (h) Write the syntax of `main()` function while using command line arguments.
- (i) What will be the output of the following code? (Explain with reason)

```
tfamily
void main() {
    int a = 3, b = 4;
    b %= 3 + 4;
    a *= a + 5;
    printf("%d %d", b, a);
}
```

- (j) What will be the output of the following code? (Explain with reason)

```
tfamily
void main() {
    int a[] = {10, 20, 30, 40, 50};
    int *p;
    p = a;
    printf("%d ", *p);
    p += 2;
    printf("%d ", *p);
}
```

Question 2

- (a) What are the characteristics of an algorithm? Draw a flowchart to print the factorial of a given number. [4]
 - (b) Explain the difference between a **while** loop and a **for** loop with a suitable example. [5]
 - (c) Explain the difference between **if-else** and **else-if** statements with a suitable example. [4]
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Question 3

- (a) Discuss about the relational operators in C with suitable examples. [5]
 - (b) Differentiate between function definition and function prototype with a suitable example. Is it possible to return more than one value from a function? Explain your answer with a suitable example. [8]
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Question 4

- (a) State the difference between automatic and external storage class variables using appropriate examples. [7]
 - (b) Write your own functions for the following operations on strings:
 - (i) Find length of a given string
 - (ii) Concatenate two strings [6]
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Question 5

- (a) Define string. Explain with a suitable example how to store and display the names of five countries. [6]
 - (b) Define pointer. Explain any three arithmetic operations that can be applied on pointer variables with suitable examples. [7]
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Question 6

- (a) Explain the syntax for opening a file in C with a suitable example. Write any two reasons for the failure of a file open operation. How do you handle this situation in your program? [7]
- (b) What do you mean by Dynamic Memory Allocation in C? Differentiate between **malloc()** and **calloc()** functions with suitable examples. [6]

Question 7

- (a) What is the role of Preprocessor in C programming? Differentiate between a Macro (with argument) and a Function with suitable examples. Out of these two, which one will you prefer and why while writing a program? [6]
- (b) Differentiate between **structure** and **union** with suitable examples. What are the situations when it would be better to use **union** instead of **structure**? Explain with a real-life example. [7]